Rohit

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Software Engineer specializing in backend systems, AI-powered products, and scalable infrastructure. Experienced building production software across distributed systems, automation, computer vision, and LLM applications with a focus on performance, reliability, and developer productivity.

Featured AI Project

WebGIS Platform, ISRO Smart India Hackathon 2024 *National Finalist*

- Built an AI-assisted WebGIS platform using Meta SAM 2 for satellite-image segmentation and geospatial analysis workflows.
- Developed the computer vision backend, REST APIs, and browser-based GIS interface under Smart India Hackathon constraints.
- National Finalist among 500+ teams, rapidly translating research ideas into production-ready features.

Experience

Alvora — Software Engineer *Oct 2025 – Present*

- Engineered across 8 Alvora platform repos (backend, frontend, scheduling solver, vehicle tracking, billing, observability); authored 406 of 2,788 commits (∼24% excluding bot/deploy accounts).
- Led backend development (Express, TypeScript, PostgreSQL) across 15 business domains; replaced Supabase with self-hosted Better Auth (Google OAuth, invite flow, role-based access) and a direct pg pool; authored 41 test suites with CI enforcement.
- Built Airbnb listing scraping and photo sync, Cloudflare R2 storage, invoice PDFs, and Google Calendar/iCal sync bridging the Timefold solver and billing microservice; hardened the API with an SSRF guard, rate limiters, and central error handling.
- Primary author of the React 19 frontend: role-based admin, client, and driver portals, 12 feature modules, i18n in 4 languages, and map-based scheduling views.
- Extended the Java Timefold scheduling solver: live-day replanning that never re-decides in-progress or committed cleanings, per-day availability windows, and travel-time replay with Monte Carlo simulation of per-person pace.
- Built live vehicle tracking (Traccar, Redis) with PR CI and end-to-end tests; stood up Loki/Grafana observability with 6 dashboards as code and 13 Telegram alerts; stabilized the FastAPI billing service with a 480-test suite.

Alvora — Software Engineer Intern *Aug 2025 – Sep 2025*

- Built a tiered scraping pipeline processing ∼3,000 listings/day: direct JSON API first (∼30 KB), Playwright browser fallback (∼2 MB), reducing per-property bandwidth by 98.5%.
- Authored 112 of 113 commits across the ingestion service; implemented batch scheduling, IP rotation, health monitoring, and a FastAPI + PostgreSQL storage pipeline.
- Engineered the data collection infrastructure used to accumulate months of property data for downstream machine learning model training and evaluation.

Celebrare — Python Developer Intern *Nov 2024 – Apr 2025*

- Evaluated and benchmarked multiple face-recognition architectures across production datasets, measuring latency, throughput, and accuracy.
- Optimized recognition quality through clustering-based experimentation and preprocessing techniques.
- Rewrote the production pipeline from Python to C++, achieving a 5–8× speedup and reducing average processing latency by 91%.

Projects

rb (Repository Brain) — Code-Intelligence Engine *Rust · PostgreSQL · Apache AGE · Tree-sitter · Docker*

- Built a deterministic Rust CLI and HTTP service (12 crates) that indexes Git repos into a queryable code graph (functions, endpoints, DB tables, DI wiring), answering caller, impact, and request-path queries with file:line evidence across 6 languages.
- Ran SCIP indexers in sandboxed Docker behind an HTTPS allow-list proxy with a Tree-sitter fallback; stored graphs in Postgres + Apache AGE and shipped an `rb check` architecture gate for CI.
- Benchmarked against independent answer keys: 122 of 130 judged edges correct; 88.7% precision and 75.8% recall on endpoints; stress-tested over 18 fixtures and 13 public repos.

Kdoc — Distributed PDF Processing Pipeline *Go · Redis Streams · SQLite · Prometheus · Docker*

- Built a Go API and scalable workers on Redis Streams consumer groups with admission control, distributed locks, retries, a replayable dead-letter queue, and stuck-job recovery.
- Load-tested with 600 simulated devices at ∼700 docs/s (0.65s avg latency); traced saturation at 10k–30k docs to SQLite write serialization.
- Wrote a race-clean 108-test suite that caught 3 production bugs; added 11 Prometheus metrics and 4 alert rules.

Mailsync *Node.js · IMAP · Elasticsearch · Qdrant · Gemini*

- Built an AI-powered email intelligence platform that continuously synchronized mailboxes through persistent IMAP connections.
- Implemented semantic search using Elasticsearch and Qdrant with retrieval-augmented generation for natural-language querying.
- Designed scalable background workers, incremental synchronization, and fault-tolerant indexing pipelines.

Skills

Languages	Python, Go, TypeScript, JavaScript, Java, C/C++
Frontend	React, Next.js, TailwindCSS, HTML/CSS
Backend	Node.js, Express, FastAPI, Flask, REST APIs, Microservices, Timefold Solver
Databases	PostgreSQL, Redis, SQLite, Supabase, Elasticsearch, Qdrant
AI	LLM Applications, RAG, Prompt Engineering, OpenCV, SAM 2, Computer Vision
Tools	Git, Docker, CI Pipelines, Grafana, Loki/Promtail, Playwright, Claude Code, Cursor
Core Skills	Distributed Systems, Automation, Performance Optimization, Scalable Backend Systems, AI Integration